WEEK 2

FROM TURING TO BUSINESS OPPORTUNITY

Chapter 2: Progressing from Turing to Deep Learning
+ Course Extension: From Capability to Business Value

CHAPTER 2 CAPSTONE MILESTONE 1

Thursday • September 3, 2026

BRIDGE

Week 1 gave us the vocabulary. Week 2 explains the progression.

1 WEEK 1 — WHAT IS AI?

- AI ≠ automation ≠ analytics
- Predictive, generative, and agentic modes
- Value appears when a workflow or decision improves
- Risk and human oversight matter

2 WEEK 2 — HOW DID WE GET HERE?

- Turing and early AI ideas
- Learning from examples
- AI winters and specialized AI
- Deep learning
- Copilot in knowledge work
- Business opportunity framing

The story moves from capability → application → business judgment.

TEXTBOOK ALIGNMENT

Chapter 2 — every official section is visible in today's lecture

We cover the chapter at an understandable level, then extend it into our graduate business capstone.

2.1	Tracing the Evolution of AI	Slides 7–8
2.2	Turing Test and Intelligence	Slides 9–10
2.3	Applying Machine Learning Innovations	Slides 11–14
2.4	Illustrating Deep Learning	Slides 16–18
2.5	Deep Learning in Microsoft Office with Copilot	Slides 20–21
2.6	Developing PowerPoint Slides using Copilot	Slides 21–23
2.7	AI's Impact on Business and the Economy	Slide 24

COURSE EXTENSION → Problem framing • capability fit • KPI • ROI • Capstone Milestone 1

WEEK 2 OUTCOMES

By the end of class, you should be able to explain and apply the chapter

1 TRACE
Explain the progression from Turing → learning systems → deep learning → Copilot.

2 DISTINGUISH
Separate rule-based logic, machine learning, and deep learning at an intuitive level.

3 EXPLAIN
Describe what the Turing test shows — and what it does not prove.

4 USE
Write a structured business prompt using the textbook's four prompt components.

5 FRAME
Turn a business pain point into a measurable AI opportunity for the capstone.

AI NOW

Three current signals — capability, adoption, governance

A recurring 5-minute ritual: classify the headline before discussing the hype.

1 CAPABILITY

OpenAI said an upcoming model required stronger safeguards because of advanced cybersecurity capabilities.

Question: What new controls become necessary as capability rises?

2 ADOPTION

Enterprise AI is moving from “answering” toward “doing” as organizations use more agentic workflows.

Question: What changes when AI can execute work, not only advise?

3 GOVERNANCE

The EU AI Act is now in a major implementation phase, making AI governance an operational business issue.

Question: Why does governance often follow capability and adoption?

PART I • CHAPTER 2

HOW AI BECAME CAPABLE

We are building the Chapter 2 mental model — not memorizing a museum of dates.

EVOLUTION

TEXTBOOK • 2.1 EVOLUTION

AI progressed in waves — not in a straight line

- **1950** — Turing asks how we might test machine intelligence
- **1956** — Dartmouth workshop helps establish AI as a field
- **1950s–60s** — Perceptron + early machine-learning ideas
- **1970s / late 1980s** — AI winters: expectations outrun capability
- **1980s–90s** — Vision, language, and robotics become more specialized
- **2010s** — Big data + GPUs accelerate deep learning
- **2020s** — Generative AI and copilots enter everyday workflows

Manager takeaway: capability rises when ideas, data, and computing infrastructure finally line up.

ORIGINS

TEXTBOOK • 2.1–2.2

Two ideas shaped the early conversation: a field and a test

1 DARTMOUTH • 1956

Researchers gathered around the idea that aspects of intelligence might be described precisely enough for machines to simulate.

Why it matters: AI became a distinct research agenda.

2 TURING • IMITATION GAME

A human judge communicates through text with a person and a machine. If the judge cannot reliably tell which is which, the machine has succeeded at imitation.

Why it matters: behavior became testable.

TURING TEST

TEXTBOOK • 2.2 TURING TEST

Can a machine imitate human conversation well enough to fool a judge?

JUDGE

Asks questions through text. Does not know which respondent is human.

→

HUMAN

Answers naturally from human knowledge and experience.

→

MACHINE

Attempts to produce responses that appear human-like.

PASSING THE TEST = convincing conversational behavior under the test conditions

IMPORTANT DISTINCTION

TEXTBOOK • 2.2 INTELLIGENCE

Human-like output is not the same as human-like understanding

✓ WHAT THE TEST CAN SHOW

- Conversational imitation
- Behavioral performance
- Ability to respond flexibly
- A useful benchmark for interaction

! WHAT IT DOES NOT DIRECTLY PROVE

- Human consciousness
- General intelligence
- Reliable factual knowledge
- Human-like reasoning in every domain

Business lesson: evaluate the capability you need — not the impression of intelligence.

MACHINE LEARNING

TEXTBOOK • 2.3 MACHINE LEARNING

The key shift: from rules written by people to patterns learned from examples

A RULE-BASED SYSTEM

Human writes the logic.

IF invoice > $10,000
THEN route to manager.

Best when rules are stable, explicit, and auditable.

B MACHINE LEARNING

Model sees many examples and adjusts internal parameters to capture useful patterns.

Example: learn which customers are likely to churn.

Best when patterns are too complex for simple rules.

EARLY LEARNING IDEAS

TEXTBOOK • 2.3 MACHINE LEARNING

Two textbook examples: learning through experience and learning weights

1 ARTHUR SAMUEL • CHECKERS

An early program improved its play through experience rather than relying only on a fixed list of moves.

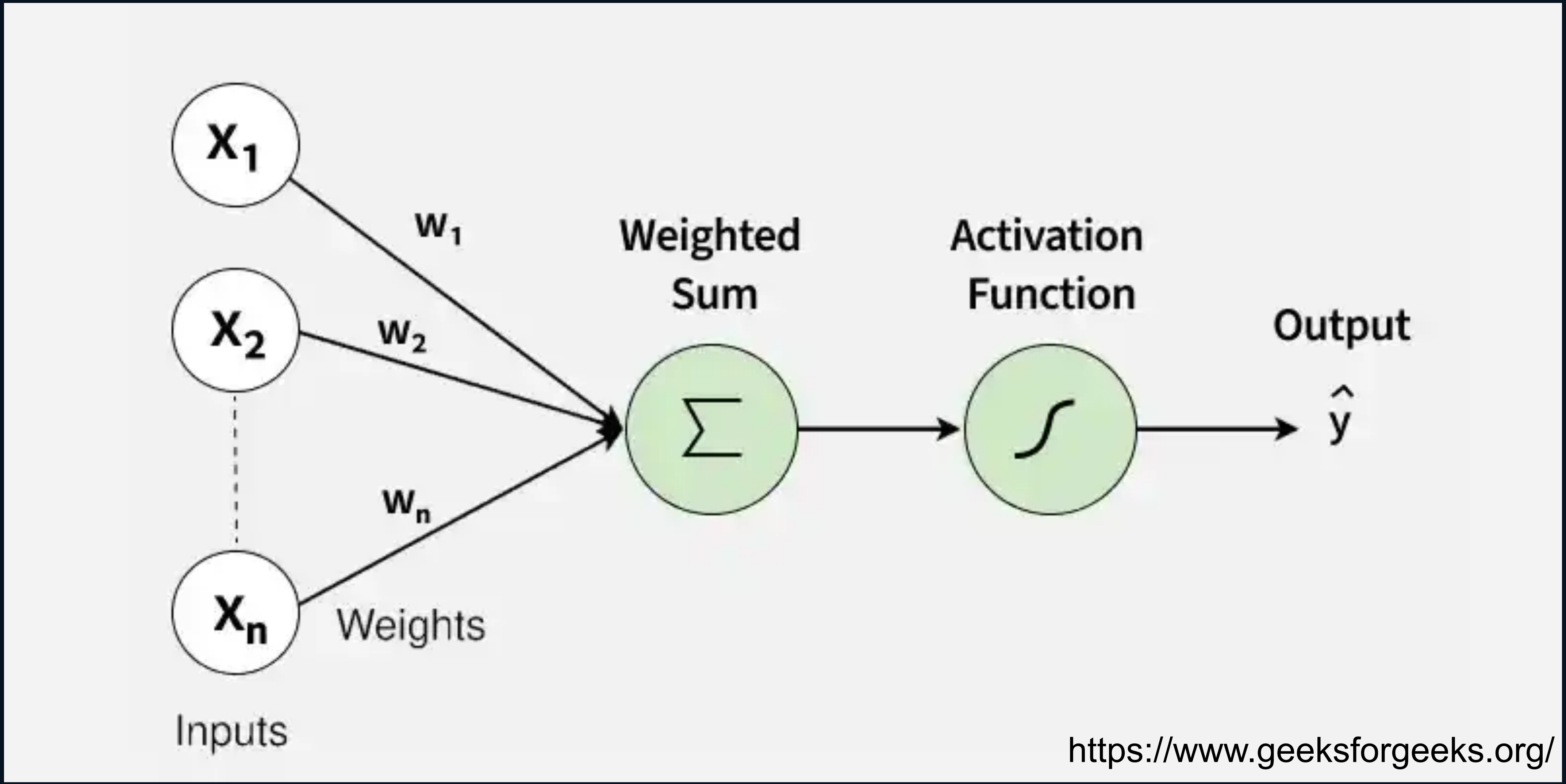
Concept to remember: a system can improve based on examples and feedback.

2 PERCEPTRON • EARLY CLASSIFIER

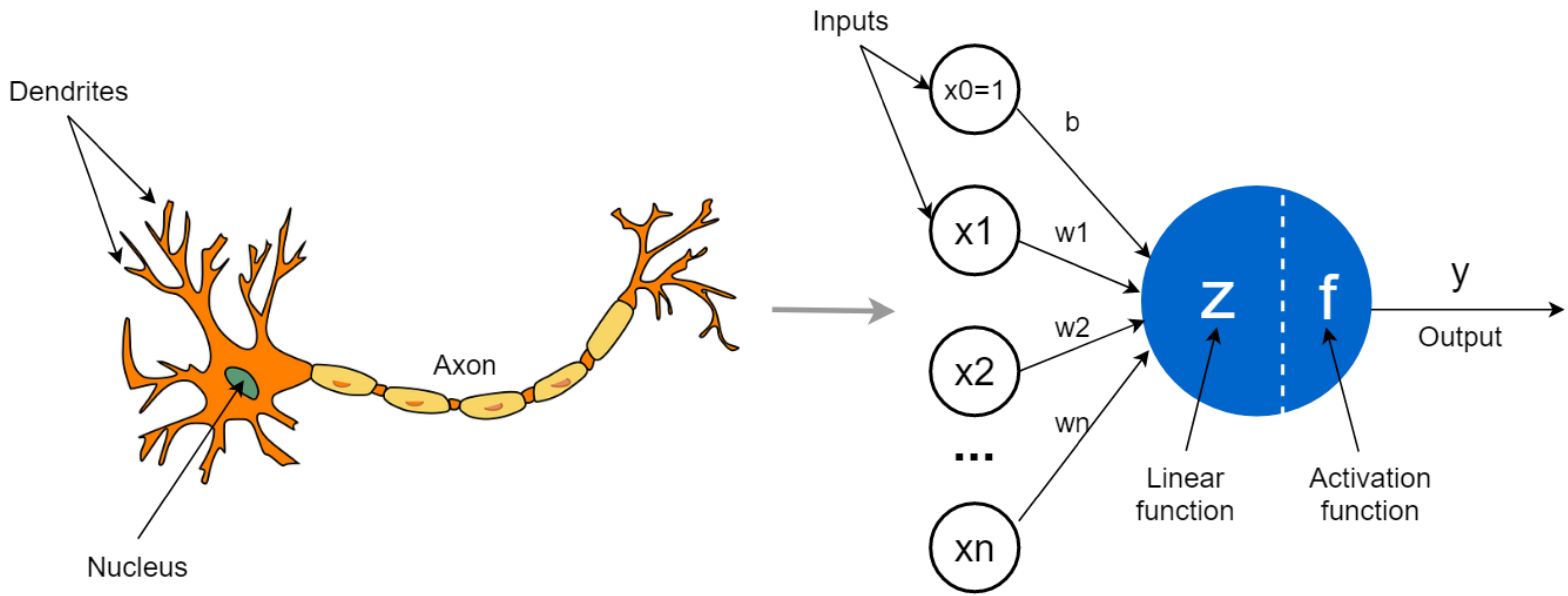
An early artificial-neuron model combined inputs using adjustable weights and changed those weights when it made errors.

Concept to remember: “learning” can mean adjusting parameters.

No equations today. The business idea is simple: examples + feedback → better classification.



X1
X2
Xn
w1
w2
wn
Weights
Inputs
Weighted Sum
Σ
Activation Function
Output
ŷ
https://www.geeksforgeeks.org/



https://towardsdatascience.com/

AI WINTERS

TEXTBOOK • 2.3 AI WINTERS

When expectations outran technology, funding and enthusiasm fell

1 BIG PROMISES

Researchers and investors expect rapid breakthroughs.

2 LIMITS EXPOSED

Compute, data, algorithms, and real-world performance fall short.

3 INTEREST FALLS

Funding and attention decline — an “AI winter.”

4 RESEARCH CONTINUES

Work persists until new enablers make progress practical again.

Executive lesson: distinguish durable capability from temporary excitement.

SPECIALIZED AI

TEXTBOOK • 2.3 SPECIALIZED AI

Vision, language, and robotics advanced by focusing on specific tasks

1 COMPUTER VISION

Works with images and video.

Examples:
- detect defects
- recognize objects
- interpret medical images

2 LANGUAGE / NLP

Works with human language.

Examples:
- classify text
- understand speech
- summarize content

3 ROBOTICS

Combines sensing, decision-making, and physical action.

Examples:
- assembly
- inspection
- warehouse movement

Preview only: later chapters revisit these areas in much greater depth.

CHAPTER 2.4 • ILLUSTRATING DEEP LEARNING

DEEP LEARNING

A machine-learning approach that learns increasingly useful representations through multiple layers.

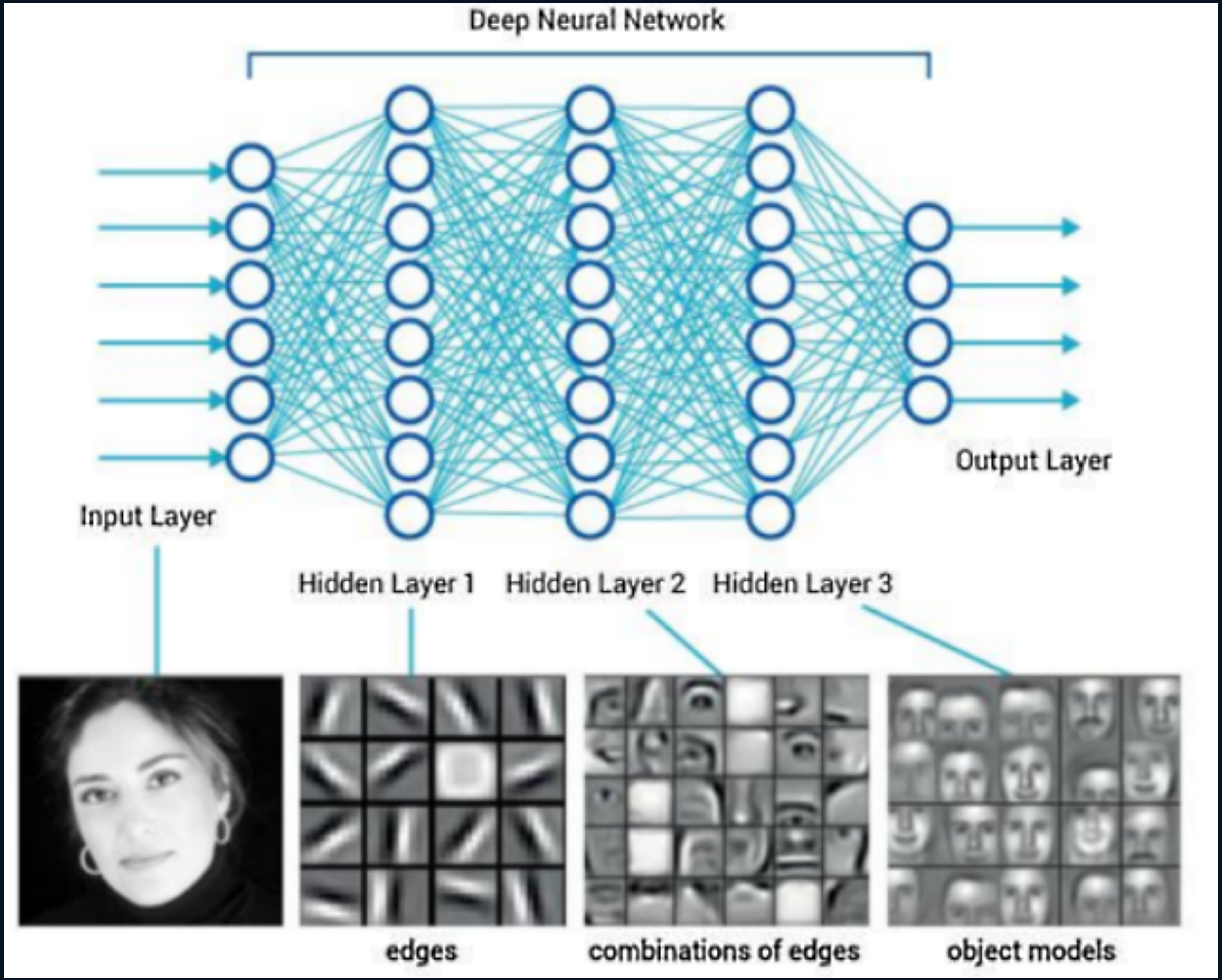
DEEP LEARNING

TEXTBOOK • 2.4 DEEP LEARNING

A simple three-part mental model: input → hidden processing → output

INPUT LAYER		HIDDEN LAYERS		OUTPUT LAYER
Receives raw information. Image pixels Text tokens Audio features Sensor readings	→	Learn useful internal patterns. Edges → shapes Words → meaning Signals → higher-level features	→	Produces the task result. Class Prediction Description Recommendation

“Deep” means many learned layers — real systems can have far more than this simplified diagram.



https://iq.opengenus.org/hidden-layers/

WHY DEEP LEARNING ACCELERATED

TEXTBOOK • 2.4 DEEP LEARNING

Three enablers converged: data, computing power, and better architectures

1 BIG DATA

Digital systems created huge collections of text, images, video, transactions, and sensor data for learning.

2 GPUs + COMPUTE

GPUs can perform many numerical operations in parallel, making neural-network training dramatically faster.

3 BETTER ARCHITECTURES

Improved neural-network designs — including attention mechanisms — made it easier to learn complex relationships.

We revisit attention and transformers later. Today, remember why capability jumped.

WHAT DEEP LEARNING ENABLED

TEXTBOOK • 2.4 APPLICATIONS

The same approach supports very different business applications

1 IMAGE RECOGNITION

defects • medical images • visual search

2 SPEECH + LANGUAGE

voice assistants • transcription • translation

3 PERSONALIZATION

recommendations • ranking • next-best action

4 AUTONOMOUS SYSTEMS

vehicles • robots • game/virtual environments

Same technology family; different task, data, risk, and business value.

CHAPTER 2.5–2.6 • COPILOT + POWERPOINT

AI ENTERS EVERYDAY KNOWLEDGE WORK

Chapter 2 uses Microsoft Copilot to show what happens when deep-learning-based AI is embedded directly into familiar productivity tools.

COPILOT IN WORD

TEXTBOOK • 2.5 COPILOT IN WORD

The textbook shows AI assisting the writing workflow — not replacing judgment

BRAINSTORM

Generate ideas or a rough first draft.

REWRITE

Change tone, clarity, length, or emphasis.

TRANSFORM

Turn text into a table or another structure.

ASK ABOUT THE DOCUMENT

Summarize or answer questions about existing content.

Textbook workflow: prompt → review the draft → refine → keep / replace only after human judgment.

TEXTBOOK • 2.6 POWERPOINT COPILOT

COPILOT IN POWERPOINT

AI can generate a first deck, add slides, and draft speaker notes — but it still needs an editor

1 CREATE

Start from a topic or source and generate a draft presentation structure.

2 REFINE

Ask for a missing slide, reorganize content, or change the emphasis.

3 REVIEW

Check facts, slide logic, audience fit, speaker notes, and visual quality before presenting.

TEXTBOOK • PROMPT ALONGS

PROMPT ENGINEERING

Use the textbook’s four components: Goal + Context + Expectations + Source

GOAL	What do you want the AI to accomplish?	**Create a 6-slide executive briefing.**
CONTEXT	Who is the audience and situation?	**Regional sales leaders reviewing Q2 performance.**
EXPECTATIONS	What should the output look like?	**Concise, professional, include risks and next steps.**
SOURCE	What evidence should it use?	**Use the supplied Q2 report; do not invent numbers.**

Prompting is business specification: clearer inputs make the output easier to evaluate.

HUMAN REVIEW

TEXTBOOK • COPILOT SECURITY + REVIEW

A polished AI draft is still a draft

FACTS

Are names, numbers, claims, and dates correct?

SOURCE

Did the model use the evidence you intended?

OMISSIONS

What important context is missing?

AUDIENCE

Is the tone and level appropriate?

PRIVACY

Did you avoid confidential or sensitive data?

UH rule of thumb: use approved tools, protect sensitive data, and verify AI output before it influences work.

TEXTBOOK • 2.7 BUSINESS + ECONOMY

BUSINESS IMPACT

AI matters economically when it changes work, decisions, customers, or products

1 AUTOMATE

Reduce repetitive effort.

Examples:
document processing
routing
first-draft work

2 AUGMENT

Improve human judgment or capacity.

Examples:
forecast support
risk review
analyst assistance

3 PERSONALIZE

Use data to tailor decisions at scale.

Examples:
recommendations
offers
service

4 REDESIGN

Change the workflow itself.

Examples:
AI-assisted process
new roles
new products

The management question is not "Do we have AI?" It is "What changed — and how will we measure it?"

MINDTAP-STYLE CHECKPOINT

CHAPTER 2 CHECKPOINT

Two statements are true. One is false. Which one?

A A

The Turing test evaluates whether conversational behavior can appear human-like under the test conditions.

B B

Machine learning requires a programmer to write a separate explicit rule for every possible case.

C C

Deep learning is a form of machine learning that can use multiple layers to learn complex representations.

Vote first. Then defend your answer in one sentence.

CHAPTER 2 SYNTHESIS

CHAPTER 2 SYNTHESIS

The chapter’s story in one picture

1 QUESTION

Can machines exhibit intelligent behavior?

Turing

→

2 LEARN

Can systems improve from examples?

ML + perceptron

→

3 SCALE

Can layered models learn complex patterns?

Deep learning

→

4 EMBED

Can AI assist inside everyday work?

Copilot

→

5 VALUE

Does capability improve a real business outcome?

Management

That last box is where our course goes beyond the textbook chapter.

PART II • COURSE EXTENSION + CAPSTONE

FROM CAPABILITY TO BUSINESS OPPORTUNITY

Knowing that AI can do something is different from proving that an organization should use it.

PROBLEM FIRST

COURSE EXTENSION • CAPSTONE

Do not start with the tool. Start with the business pain.

✕ TOOL-FIRST — WEAK

"Where can we use AI?"
"Let's build a chatbot."
"Can we add an agent?"

Problem: the technology is chosen before the need is defined.

✓ PROBLEM-FIRST — STRONG

"Which important decision or workflow is slow, costly, inconsistent, or difficult — and can we measure improvement?"

Business pain → decision/workflow → capability.

STRONG PROBLEM STATEMENT

COURSE EXTENSION • CAPSTONE

Five pieces turn a vague complaint into a measurable opportunity

PAIN

What is not working?

Agents spend too long finding policy answers.

ACTOR

Who experiences it?

Customer-service agents and customers.

WORKFLOW

What decision/ process?

Find the correct refund policy and respond.

BASELINE

How does it perform now?

Average search time = 11 minutes.

KPI

What should improve?

Target <4 minutes; no increase in policy errors.

Strong version: “Can we reduce policy-search time from 11 to <4 minutes without increasing incorrect policy guidance?”

CAPABILITY FIT

COURSE EXTENSION • CAPSTONE

Match the business task to the simplest useful capability

CLASSIFY
What category?
spam • intent • defect

PREDICT
What is likely?
churn • demand • risk

RECOMMEND
What next/best?
product • action • offer

GENERATE
Create or transform
text • image • summary

OPTIMIZE
Best allocation?
route • schedule • price

DETECT
What is unusual?
fraud • anomaly • failure

Sometimes the best solution is a rule, workflow redesign, search system, or better data — not AI.

AI SUITABILITY

COURSE EXTENSION • CAPSTONE

Five green lights before you invest

1	**REPEATED / IMPORTANT**	The problem happens often enough or matters enough.
2	**PATTERN / UNCERTAINTY**	There is something to learn, infer, retrieve, or generate.
3	**EVIDENCE / DATA**	Credible information exists and can be accessed responsibly.
4	**MEASURABLE OUTCOME**	A KPI or decision-quality measure can show improvement.
5	**CONTROLLABLE RISK**	Errors can be monitored, reviewed, reversed, or bounded.

If one light is red, investigate the non-AI alternative before building a model.

MEASURE BEFORE BUILDING

COURSE EXTENSION • CAPSTONE

Baseline + KPI + guardrail + ROI make the opportunity testable

1 BASELINE

Current performance.

Example:
11 min average search time
14% escalation rate

2 KPI / TARGET

What should improve?

Target:
<4 min average search time

3 GUARDRAIL

What must not get worse?

Example:
policy error rate ≤ current level

4 ROI PREVIEW

$$\frac{\text{Benefit} - \text{total cost}}{\text{Total cost}}$$

Include software, integration, review, training, monitoring.

A technically impressive prototype is not enough. We need evidence that the organization is better off.

AI OPPORTUNITY CLINIC

CAPSTONE MILESTONE 1

produce a defensible candidate, not a finished solution

1 CHOOSE	One recurring decision or workflow.
2 FRAME	Pain • actor • workflow • baseline.
3 MATCH	Primary AI capability + non-AI alternative.
4 TEST	Five green lights + evidence/data source.
5 DEFINE	KPI • target • one risk/guardrail.

Milestone 1 submission: 1–2 pages + one-slide summary. Approval gate before prototype work.

EXECUTIVE TAKEAWAY

Leave Week 2 with five rules

1. AI progressed in waves — capability depends on ideas, data, and compute.
2. The Turing test evaluates behavior, not guaranteed understanding.
3. Machine learning learns patterns; deep learning builds layered representations.
4. Copilot shows how AI enters everyday work — but prompts and verification still matter.
5. A business opportunity starts with a measurable problem, not an AI tool.

BEFORE NEXT CLASS → Complete selected MindTap Chapter 2 work + submit individual Opportunity Proposal

NEXT WEEK → Data Foundations & the AI Factory